

Functional Lost Pet Alert System

Overview — lost_found module

PawGuard Platform | pawguard-backend (FastAPI + PostgreSQL + Redis + ARQ) | Prepared for Team Review

1. Purpose

Lets an owner report a lost pet, lets any member of the public report a pet they've found, and automatically matches the two using a weighted scoring algorithm — then routes a verified match through an ownership-claim review before reuniting pet and owner.

2. Core Entities

Table	Purpose
lost_reports	One row per lost-pet report (species, breed, color, collar, last-seen location + GPS, lost_at, status, photo)
found_reports	One row per found-pet sighting (observed breed/color/collar, location + GPS, found_at, status, photo)
report_matches	System-generated pairing of a lost_report × found_report with a confidence_score, distance_km, temporal_gap_days, and match_reasons

3. End-to-End Flow

- Owner submits POST /lost-found/lost with pet details + last-seen address (geocoded to lat/long).
- Finder submits POST /lost-found/found with observed details + sighting location.
- On each new report, the matching engine (service._run_matching_for_lost / _run_matching_for_found) scores it against all open reports of the opposite type and stores any resulting report_matches row.
- Owner (or admin) triggers POST /lost-found/lost/{id}/broadcast to push the alert out (see Broadcast Alerts doc).
- Either party can view candidate matches (GET /lost-found/lost/{id}/matches) and submit an ownership claim with proof (POST /lost-found/matches/{id}/claim).
- Staff reviews the claim (POST /lost-found/matches/{id}/claim/review) — approval resolves both reports as RESOLVED.

4. Related Detailed Docs

- Broadcast Alerts — how a lost-pet alert is delivered to nearby users
- GPS Location Pinning — how location and distance-based matching work
- QR Tag Scan Flow — how a found pet's safety tag speeds up reunification

5. Privacy Safeguard

List/detail endpoints mask the reporter's exact identity (router._mask_reporter_identity) — the public sees only what's needed to help, contact is brokered through the platform until a claim is verified.

Source: lost_found/models.py, lost_found/service.py, lost_found/router.py.